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1 nmol / 0.1 mol

Homework for Chap 12

1. Calculate the plasmon energy for n-type GaAs with carrier concentrations of 10^{10}, 10^{15} and 10^{20} cm^3.
2. Calculate the density of electrons at which $\hbar\omega_{PL} = \hbar\Omega_{LO}$ for ZnO and InAs, here $\hbar\omega_{PL}$ and $\hbar\Omega_{LO}$ are the plasmon energy and longitudinal optical phonon energy, repectively. Up to which temperatures are the electron gases degenerate?

1. $\omega_p = \sqrt{\frac{ne^2}{\varepsilon_0 m_e^* \varepsilon_\infty}}$

$m_e^* = 0.067 m_e$

$n = 10^{10}\ cm^{-3}$ $\omega_p = 6.60\times10^9\ rad/s$. $E_1 = \hbar\omega_p = 4.35\ \mu eV$

$n = 10^{15}\ cm^{-3}$ $\omega_p = 2.09\times10^{12}\ rad/s$ $E_2 = \hbar\omega_p = 1.37m\mu eV$

$n = 10^{20}\ cm^{-3}$ $\omega_p = 6.60\times10^{14}\ rad/s$ $E_3 = \hbar\omega_p = 4.35\times10^{-1}\ eV$

2. $\omega_p = \sqrt{\frac{ne^2}{\varepsilon_0 \varepsilon_\infty m_e^*}} = \Omega_{LO}$, $E_F = \frac{\hbar^2}{2m^*}(3\pi^2 n)^{\frac{2}{3}}$

① ZnO: $\varepsilon_\infty = 3.7$. $m_e^* = 0.24 m_e$ $\hbar\Omega_{LO} = 0.072eV$

$\Rightarrow n_{ZnO} = 3.34\times10^{24}\ m^{-3} = 3.34\times10^{18}\ cm^{-3}$

$\Rightarrow E_F = 0.034eV$, $T_F = \frac{E_{F(ZnO)}}{k_B} = \underline{394K}$

② InAs $\varepsilon_\infty = 12.3$. $\hbar\Omega_{LO} = 0.029eV$ $m_e^* = 0.023 m_0$

$\Rightarrow n_{InAs} = 1.73\times10^{23}\ m^{-3} = 1.73\times10^{17}\ cm^{-3}$

$\Rightarrow E_F = 0.049eV$, $T_F = \frac{E_F}{k_B} = \underline{570K}$